

support operations in a unified system. Widely adopted by small and mid-sized enterprises, Zoho CRM is a part of Zoho's larger suite of productivity and business software.

Customer Relationship Management (CRM): A system used by organizations to manage interactions with customers, store customer data and improve sales and service efficiency.

Benefits:

- Centralized customer information
- Improved customer communication
- Better sales tracking and forecasting
- Enhanced customer satisfaction

How do businesses use CRM to manage customer relationships and sales?

- Managing customer contacts and ~~leads~~ leads
- Tracking sales opportunities
- Automatically follow-ups and emails
- Analyzing customer data for decision making

Contacts Management: Stores and organizes customers and lead information in one place

Deals: Tracks potential customers from first contact to final sale

Sales Pipeline Management: Visual view of where each deal stands in the sales process

Workflow Automation: Automates repetitive tasks

Introduction to SaaS & CRM

Software as a Service (SaaS): A business model in which customers pay to access and use cloud hosted software over the internet rather than purchasing it. It is differed from a traditional software that has to be purchased and installed locally.

Benefits

Cost Effectiveness: Reduces or eliminates initial infrastructure investment.

Scalability & Effectiveness: Users can easily scale subscriptions up or down based on business needs and demands.

Security & Reliability: SaaS offers high-level security, backups and disaster recovery capabilities.

Example: Google Docs is an example of SaaS.

Zoho CRM

Zoho CRM is a cloud-based customer relationship management platform developed by Zoho Corporation. It helps businesses manage sales, marketing and customer

What is Trailhead?

Salesforce trailhead is Salesforce's free online learning platform that provides interactive, hands-on training for Salesforce skills.

Features:

- Guided learning modules
- Hands-on practice using Trailblazer Playgrounds
- Badges and points for completed tasks

What is Trailblazer Playground?

Trailblazer Playground is a free Salesforce developer environment used to build, test and practice Salesforce applications without affecting real data.

Creating a new Trailblazer App:

Log-in to Salesforce Trailhead → Profile → Hands on org →

Create playground to get a ready to use app development environment

Introduction to Lightning App Builder

Lightning App Builder is a low-code drag and drop tool used to design Salesforce app pages and user interfaces easily.

Building a basic app in Salesforce!

→ Create an app? using Salesforce tools without modifying Schemas or databases

→ Creating a custom object

→ Adding fields and datatypes

→ Designing a user interface

→ Understanding no code/low code development

Testing and Debugging the App: Check app functionality by executing records and fixing errors using Salesforce debug tools.

Deploying: Deploying in Sandbox Environments: Move the tested app to a sandbox for safe testing before production use.

Future Improvements: Enhance the app by adding AI features, third-party integrations, and advanced automations.

Use of Salesforce PaaS by companies: Companies use Salesforce PaaS to build scalable business apps for CRM, automation, analytics and workflow management

Student Project: Attendance & Tracking App: Built a sample attendance app on Trailhead using custom objects, fields and lightning App

Demonstration of a PaaS Application

PaaS: A cloud computing model that provides developers with a fully managed platform to build, test, deploy and scale applications without managing infrastructure such as servers, storage, operating systems or networking.

Benefits of PaaS:

- No infrastructure management
- Faster development
- Scalability
- Cost effective
- Focus on coding and logic
- Ex: Salesforce Trailhead, AWS Elastic Beanstalk

Introduction to Salesforce and Trailhead: Salesforce is a CRM platform and a PaaS provider that enables organizations to build business applications using cloud-based tools.

Key features:

- Cloud based CRM
- Built-in database and security
- Low code/no code

2. Python

→ Open terminal

→ Use sudo apt install python3 python3-pip

3. Verification

→ python3 --version

As Running Python Program:

1. Create hello.py

2. Add simple print statement

3. Execute the file
→ python3 hello.py

CloudSim Simulator

CloudSim is a simulation toolkit used to model and test cloud computing environments without needing real cloud infrastructure.

1. Install Java

→ verify with "java -version"

2. Download CloudSim

→ Go to the official CloudSim repository

→ Download .zip file

→ Extract in file system

3. Set up and IDE

→ IntelliJ IDEM

4. Create a Java Project

→ Create a new Java project

→ Add CloudSim jar files

5. Import CloudSim packages

→ import org.cloudbus.cloudsim.*;

6. Initialize CloudSim

→ CloudSim.initUsers, calendar, traceFlag;

2. Create Datacenter

- Define hosts (CPU, RAM, storage)
- Wrap them into a datacenter

3. Create Broker

- Datacenter Broker broker = new DatacenterBroker("Broker")

4. Create VM

- Define RAM, CPU, bandwidth, storage

5. Create Cloudlets

- Define length, file size, output size,

6. Submit VMs and Cloudlets to Broker

- broker.submitVmList(vmlist);
- broker.submitCloudletList(cloudletList);

7. Start simulation

- CloudSim.startSimulation();
- CloudSim.stopSimulation();

8. Get results

- Check execution time, completion status, resource usage

Virtual Machine in VMware

VMware Workstation Pro is a powerful Type 2 hypervisor that enables users to run multiple, isolated operating systems.

Virtual Machine Creation:

1. VMware ~~trick~~ installation

- Install VMware software from Broadcom
- Use trial service

2. New VM setup

- Download 150 file of OS
- Assign new VM name and storage location

3. Resource Allocation

- Disk space
- RAM
- CPU
- Virtual disk

4. OS Installation

- VM password
- Install Linux as provided on the setup

Python Installation:

1. System Boot
 - Login and startup

Kubernetes Demonstration

1. Start cluster

minikube start

2. Verify cluster

kubectl get nodes

3. ~~Create~~ deployment

3. Create deployment

kubectl create deployment my-app --image=^{^ my code}~~nginx~~ my-app

4. Check pods

kubectl get pods

5. Expose deployment

kubectl expose deployment my-app --type=NodePort
--port=3000

6. Access application

minikube service my-app

7. Scale application

kubectl scale deployment my-app --replicas=3

kubectl get pods

8. Describe and deleting

kubectl describe pod ~~my-app~~ my-app

kubectl logs ~~my-app~~ my-app

9. Update deployment

```
kubectl set image deployment/my-app my-app=nginx/  
my-app:v2
```

10. Delete resources

```
kubectl delete service my-app  
kubectl delete deployment my-app
```

use my-app

Introduction to Cloud Course on AWS Educate

AWS Educate is a cloud-based learning platform that provides students with hands-on experience in cloud concepts using AWS.

Cloud course introduces us to the fundamentals of cloud computing, including on-demand resource provisioning, scalability and pay-as-you-go pricing models. The lab environment allowed us to explore core AWS services such as VPCs, storage and basic networking.

The AWS console is used to understand how cloud ~~paths~~ resources are created and managed in real time. Cloud course emphasized the advantages of cloud computing over traditional systems, such as flexibility, cost efficiency and global accessibility.

Overall, cloud courses on AWS Educate provide a practical introduction to cloud platforms and help build a foundational understanding of deploying and managing resources in a cloud environment.

Steps to access Cloud Courses on AWS Educate
1. Go to AWS Educate website

Docker Demonstration

1. Install and Verify Docker

`docker --version`

`docker run hello-world`

2. Create a simple app

Ex: Node.js / Python app

`mkdir docker-demo`

`cd docker-demo`

3. Write Dockerfile

`FROM node:18`

`WORKDIR /app`

`COPY . .`

`CMD ["node", "app.js"]`

4. Build Docker Image

`docker build -t my-app .`

5. Run container

`docker run my-app`

or

`docker run -p 3000:3000 my-app`

6. Check running containers

`docker ps`

`docker ps -a`

resource usage

Getting Started with Security on AWS Educate
AWS Educate introduces security fundamentals covering
new AWS tools and concepts

1. Understand the Shared Responsibility Model

→ AWS secures cloud infrastructure

→ Customer secures: Data

Access

Configurations

2. Learn Identity and Access Management (IAM)

→ Use AWS Identity and Access Management

→ Create: Users

Groups

Roles

→ Apply least privilege principles

3. Secure Login with Multiple-Factor Authentication

→ Enable MFA for accounts

→ Adds extra verification layer beyond password

→ Prevents unauthorized access

4. Manage Permissions with Policies

→ Attach policies to user/roles

→ Define: What actions are allowed

Which resources can be accessed

→ Example: Read-only vs full access

5. Monitor Activity and logs

→ Use AWS CloudTrail

→ Tracks :: login attempts

· API calls

· Changes in resources

→ Helps in auditing and troubleshooting

6. Protect Data with Encryption

→ Data can be encrypted :: At rest

· In transit

→ Use AWS Key Management Service for key handling

7. Network-level Security Basics

→ Use :: Security Groups (firewall rules)

· VPC settings

→ Control inbound and outbound traffic

Getting Started with Compute on AWS Educate

A main service used for computing power on AWS is EC2, Elastic Cloud Compute is a service by AWS that provides virtual servers in the cloud. It lets you select OS, ~~mem~~ RAM, storage etc. You can run applications just like on a real machine. You only pay for what you use.

Types of EC2 Instances:

- General Purpose
- Compute Optimized
- Memory Optimized
- Storage Optimized
- GPU/Accelerated Computing

Steps to Create an EC2 Instance:

1.

- 1. Login to AWS console
- Go to AWS dashboard
- Open EC2 service

2. Launch Instance

- Click Launch Instance

3. Choose AMI (Operating System)

- Eg: Amazon Linux, Ubuntu

2. Create account

Click "Join Axis Educatn"

3. Complete Profile Setup

Fill in: → name

→ college

→ areas of interest

4. Access the Learning Dashboard

It shows: → Learning Pathways

→ Available courses

→ Progress Tracking

5. Browse Cloud Courses

Navigate to Cloud Career Pathways

Select topics like: → Cloud Fundamentals

→ Networking

→ Security

→ ML

6. Enroll in a course

Select "Start Learning"

7. Start Learning Modules

Course includes: → Videos

→ Labs

→ Quizzes

Getting Started with Serverless on AWS Educate

AWS Educate introduces serverless computing from AWS where you can run code without provisioning or managing servers

1. Understand Serverless Concept

- No need to manage infrastructure
- Code runs on-demand
- Automatic scaling

2. Explore AWS Lambda

- Use AWS Lambda (JS, Python, Node.js, etc)
- Upload code (Python, Node.js, etc)
- Function runs when triggered

3. Configure Triggers

- Connect Lambda with:
 - API requests
 - Storage events

→ Common pairing is Amazon API Gateway

→ Enables building web APIs without servers

4. Manage Permissions

- Use IAM roles
- Grant Lambda only required access

5. Monitor Execution

- Use Amazon CloudWatch
- Check:

- Logs

- Errors

- Execution time

6. Understand Pricing Model

- Pay only when code runs
- No cost for idle time
- Very efficient for event-driven apps

Creating a Lambda Function in AWS:

1. Login to AWS Console
- Open AWS Dashboard
- Go to Lambda service

2. Click "Create Function"

- Choose "Author from scratch" out of options

3. Configure Basic Details

- Function name
- Runtime:
 - Python
 - Node.js
 - Java
- Runtime decides the language used to code function